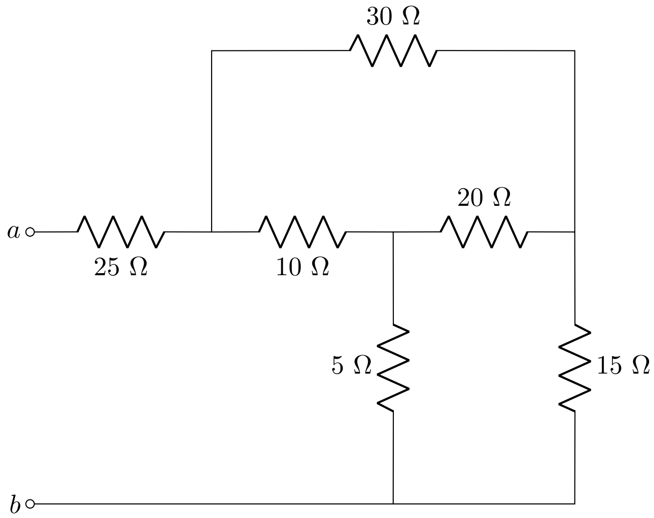
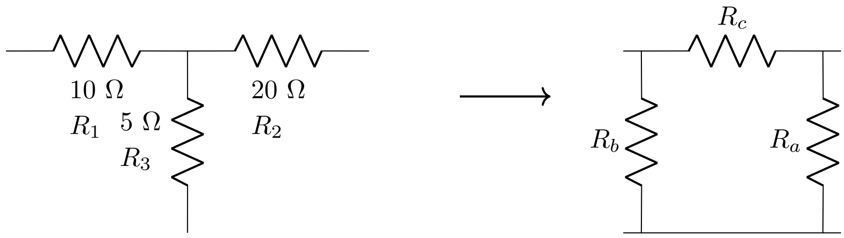
Tutorial 3

Question 2.51b

Determine the equivalent resistance across $a - b$.



Memo: Perform a y-delta conversion on the middle 3 elements

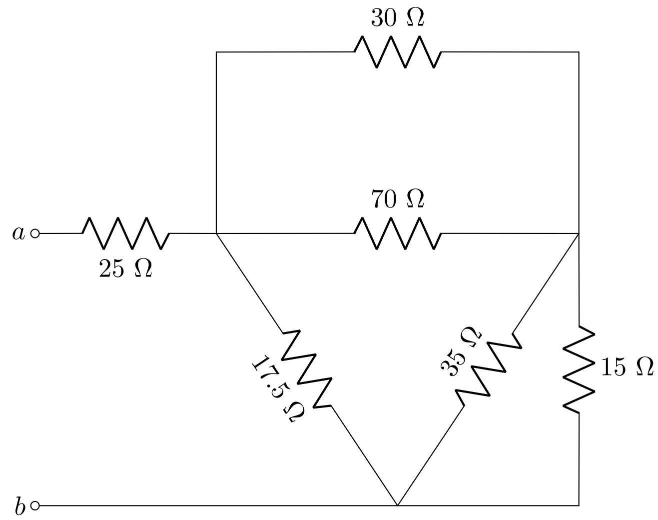


$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} = 35\ \Omega$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} = 17.5\ \Omega$$

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = 70\ \Omega$$

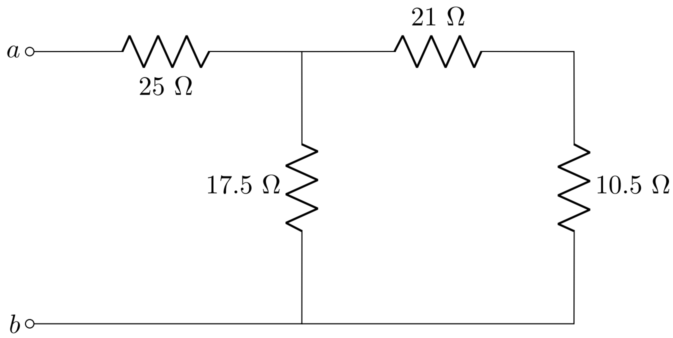
Redraw



$$70||30 = 21\ \Omega$$

$$35||15 = 10.5\ \Omega$$

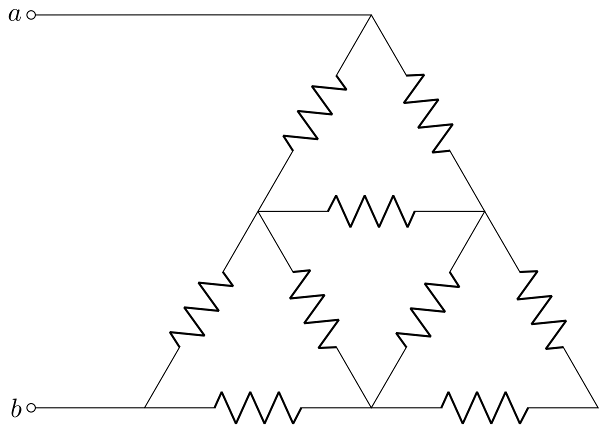
Thus



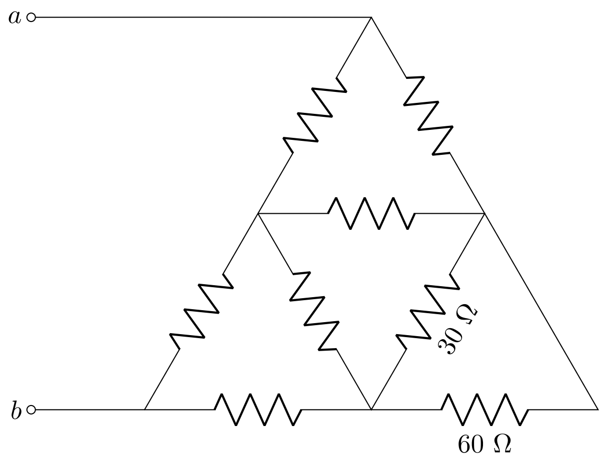
$$R_{ab} = 25 + 17.5||(21 + 10.5) = 36.25\ \Omega$$

Question 2.53b

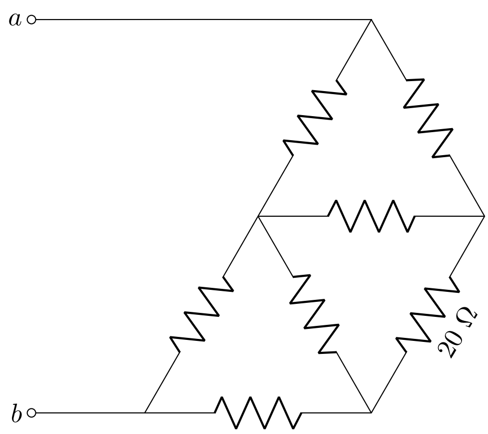
Determine the equivalent resistance across $a - b$ if all the elements are 30 Ω.



Memo: Combine the bottom right 2 series elements to get

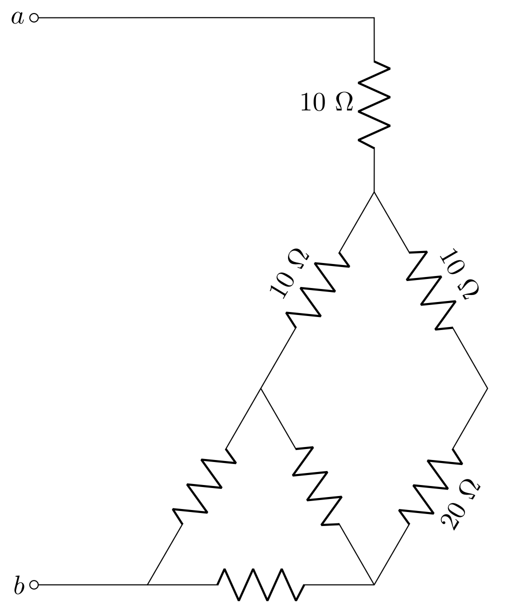


This can be combined with the 30 Ω in parallel:

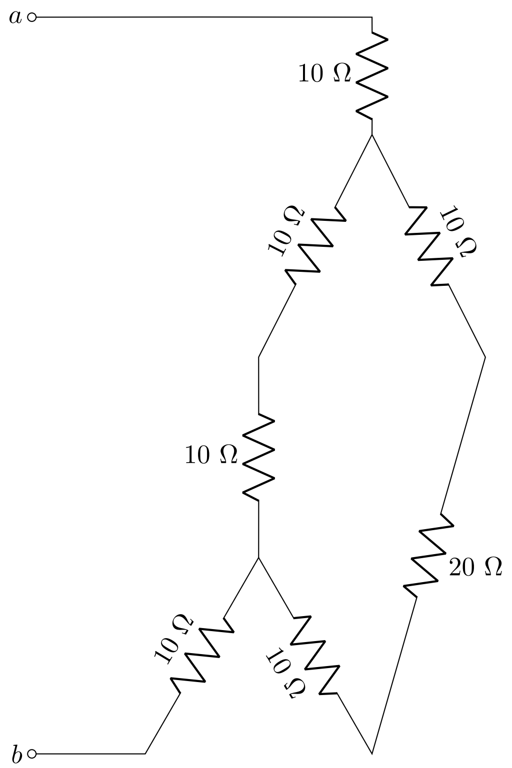


We can now convert the top triangle using y delta

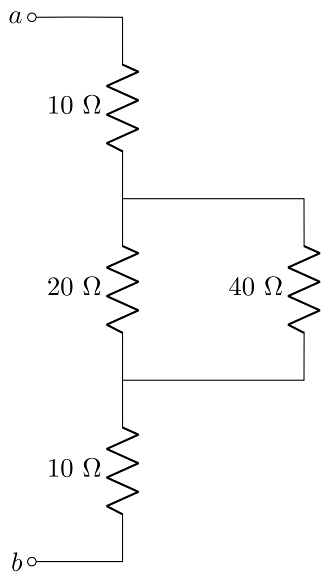
$$R_y = R_\Delta/3 = 10\ \Omega$$



Now we can do the same thing with the bottom left triangle



The 2 10s and the 20 on the right are in series as well as the two tens in the middle. This simplifies to:

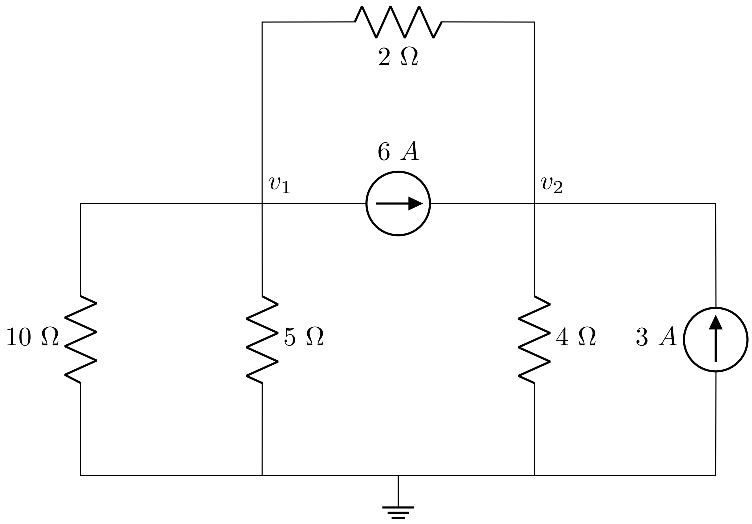


Thus:

$$R_{ab} = 10 + 20||40 + 10 = 33.33\ \Omega$$

Question 3.2

Determine the nodal voltages v_1 and v_2.



Memo:

Node v_1 assuming all currents leave node

$$\frac{v_1 - v_2}{2} + 6 + \frac{v_1}{5} + \frac{v_1}{10} = 0$$

Multiply by 10 (lowest common denominator)

$$5(v_1 - v_2) + 60 + 2v_1 + v_1 = 0$$

$$8v_1 - 5v_2 = -60 \tag{1}$$

Node v_2 assuming all currents leave node

$$\frac{v_2 - v_1}{2} - 6 + \frac{v_2}{4} - 3 = 0$$

Multiply by 4 (lowest common denominator)

$$2(v_2 - v_1) - 24 + v_2 - 12 = 0$$

$$-2v_1 + 3v_2 = 36 \tag{2}$$

Perform simultaneous equations: (1) in (2)

$$-2\frac{5v_2 - 60}{8} + 3v_2 = 36$$

$$-5v_2 + 60 + 12v_2 = 144$$

$$v_2 = 12\ V$$

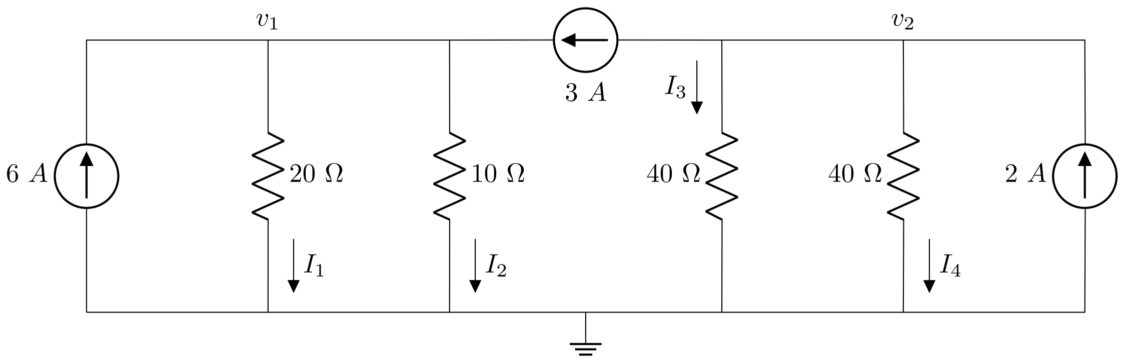
Back to (1)

$$8v_1 - 5(12) = -60$$

$$v_1 = 0\ V$$

Question 3.4

Determine currents I_1, I_2, I_3 and I_4.



Memo:
Node v_1 assuming all currents leave node

$$-6 + \frac{v_1}{20} + \frac{v_1}{10} - 3 = 0$$

Multiply by 20 (lowest common denominator)

$$-120 + v_1 + 2v_1 - 60 = 0$$

$$v_1 = 60\ V$$

Node v_2 assuming all currents leave node

$$3 + \frac{v_2}{40} + \frac{v_2}{40} - 2 = 0$$

Multiply by 40 (lowest common denominator)

$$120 + v_2 + v_2 - 80 = 0$$

$$v_2 = -20\ V$$

Calculate currents:

$$I_1 = v_1/20 = 60/20 = 3\ A$$

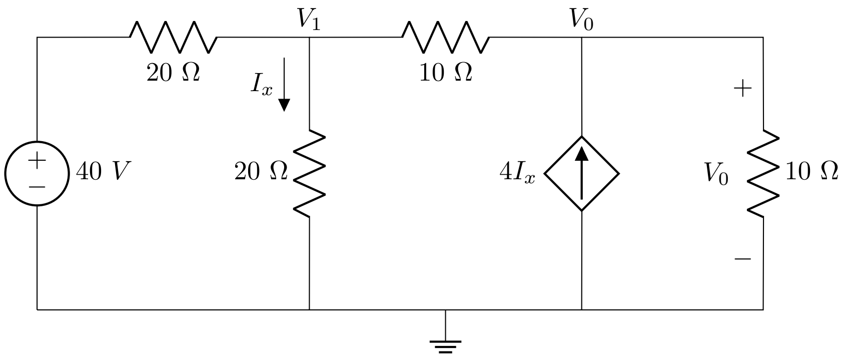
$$I_2 = v_1/10 = 60/10 = 6\ A$$

$$I_3 = v_2/40 = -20/40 = -0.5\ A$$

$$I_4 = v_2/40 = -20/40 = -0.5\ A$$

Question 3.12

Determine V_1 and V_0.



Memo:

Node V_1 assuming all currents leave node

$$\frac{V_1 - 40}{20} + \frac{V_1}{20} + \frac{V_1 - V_0}{10} = 0$$

$$V_1 - 40 + V_1 + 2V_1 - 2V_0 = 0$$

$$4V_1 - 2V_0 = 40 \tag{3}$$

Node V_0 assuming all currents leave node

$$\frac{V_0 - V_1}{10} + \frac{V_0}{10} - 4I_x = 0$$

$$I_x = \frac{V_1}{20}$$

$$\frac{V_0 - V_1}{10} + \frac{V_0}{10} - 4\frac{V_1}{20} = 0$$

$$V_0 - V_1 + V_0 - 2V_1 = 0$$

$$-3V_1 + 2V_0 = 0 \tag{4}$$

Perform simultaneous equations: (4) in (3)

$$4V_1 - 2(3/2V_1) = 40$$

$$V_1 = 40\ V$$

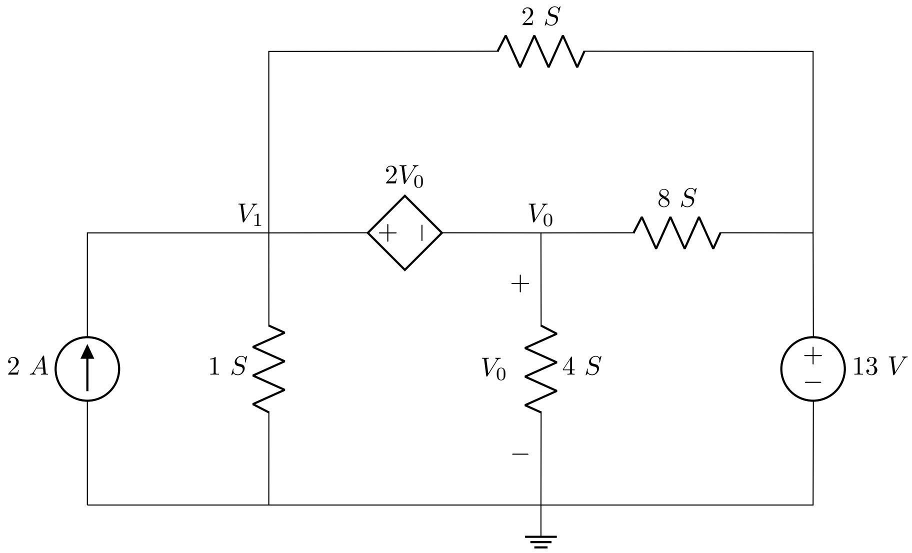
Back to (4)

$$3/2V_1 = V_0$$

$$V_0 = 60\ V$$

Question 3.16

Determine V_0.



Memo:
V_1 and V_0 form a supernode: Thus

$$V_1 - V_0 = 2V_0$$

$$V_1 - 3V_0 = 0 \tag{5}$$

Assuming all currents leave supernode

$$-2 + 1V_1 + 2(V1 - 13) + 8(V_0 - 13) + 4V_0 = 0$$

$$3V_1 + 12V_0 = -132 \tag{6}$$

Perform simultaneous equations: (5) in (6)

$$3(3V_0) + 12V_0 = -132$$

$$V_0 = 6.286\ V$$

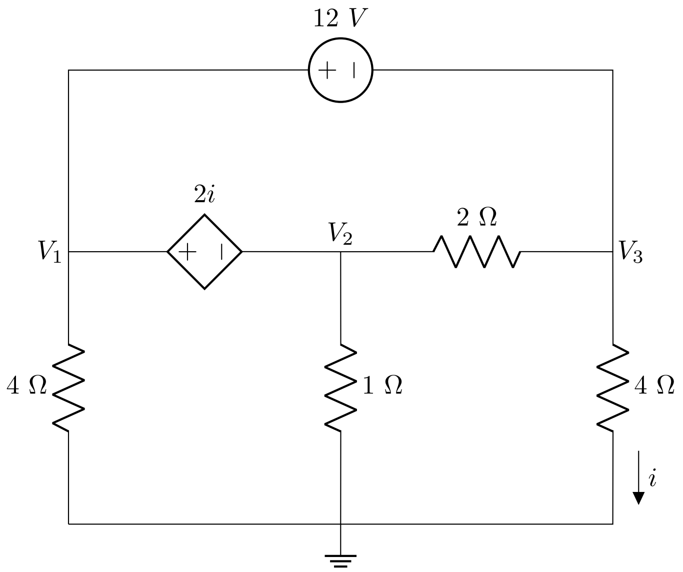
Back to (5)

$$V_1 = 3V_0$$

$$V_1 = 18.857\ V$$

Question 3.20

Determine V_1, V_2 and V_3.



Memo:

V_1 and V_2 and V_3 form a supernode with 2 equations within:

$$V_1 - V_2 = 2i$$

$$i = V_3/4$$

$$V_1 - V_2 - 0.5V_3 = 0 \tag{7}$$

$$V_1 - V_3 = 12 \tag{8}$$

Assuming all currents leave supernode

$$\frac{V_1}{4} + \frac{V_2}{1} + \frac{V_2 - V_3}{2} + \frac{V_3 - V_2}{2} + \frac{V_3}{4} = 0$$

Multiply by 4

$$V_1 + 4V_2 + V_3 = 0 \tag{9}$$

Perform simultaneous equations: (8) in (7)

$$V_1 - V_2 - 0.5(V_1 - 12) = 0$$

$$0.5V_1 - V_2 = -6 \tag{10}$$

(8) in (9)

$$V_1 + 4V_2 + (V1 - 12) = 0$$

$$2V_1 + 4V_2 = 12 \tag{11}$$

(10) in (11)

$$2V_1 + 4(0.5V_1 + 6) = 12$$

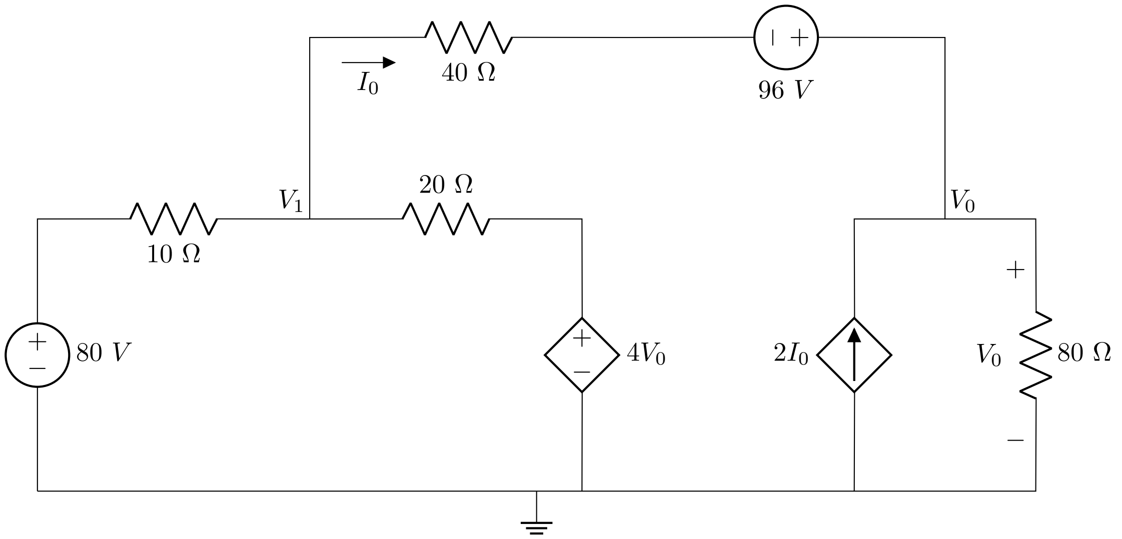
$$V_1 = -3\ V$$

$$V_2 = 4.5\ V$$

$$V_3 = -15\ V$$

Question 3.30

Determine V_0 and I_0.



Memo:
Node V_1 assuming all currents leave node

$$\frac{V_1 - 80}{10} + \frac{V_1 - 4V_0}{20} + \frac{V_1 - V_0 + 96}{40} = 0$$

$$4V_1 - 320 + 2V_1 - 8V_0 + V_1 - V_0 + 96 = 0$$

$$7V_1 - 9V_0 = 224 \tag{12}$$

Node V_0 assuming all currents leave node

$$-2I_0 + \frac{V_0 - V_1 - 96}{40} + \frac{V_0}{80} = 0$$

$$I_0 = \frac{V_1 - V_0 + 96}{40}$$

$$-2\frac{V_1 - V_0 + 96}{40} + \frac{V_0 - V_1 - 96}{40} + \frac{V_0}{80} = 0$$

$$-4V_1 + 4V_0 - 4(96) - 2V_1 + 2V_0 - 2(96) + V_0 = 0$$

$$-6V_1 + 7V_0 = 576 \tag{13}$$

Perform simultaneous equations: (12) in (13)

$$-6\left(\frac{224 + 9V_0}{7}\right) + 7V_0 = 576$$

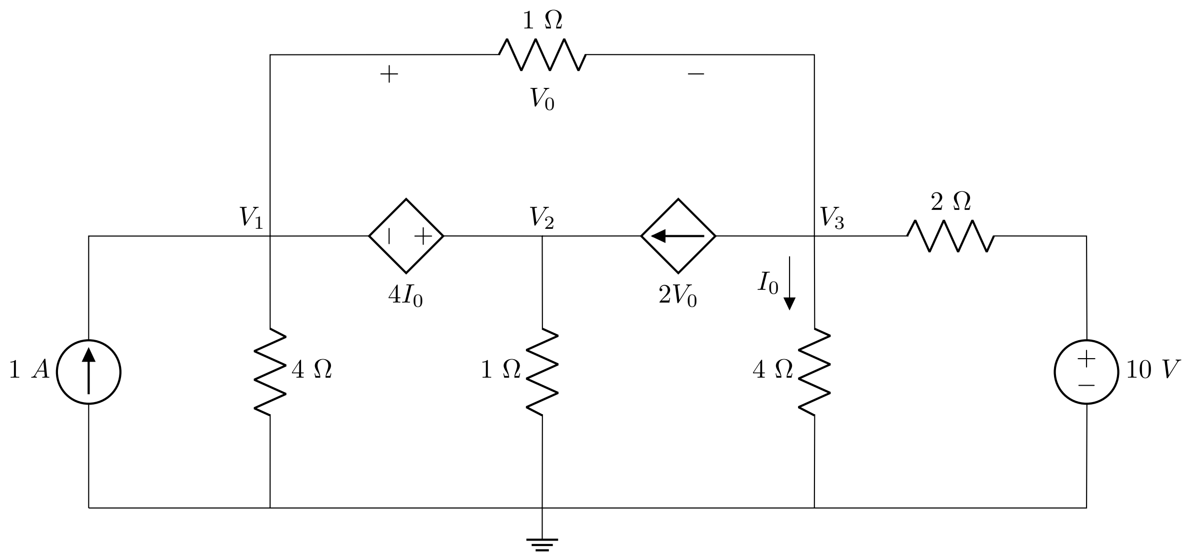
$$V_0 = -1.0752\ kV$$

$$V_1 = -1350.4\ V$$

$$I_0 = \frac{V_1 - V_0 + 96}{40} = -4.48\ A$$

Question 3.31

Determine V_0.



Memo:

V_1 and V_2 form a supernode: Thus

$$V_2 - V_1 = 4I_0$$
$$I_0 = V_3/4$$
$$-V_1 + V_2 - V_3 = 0 \tag{14}$$

Assuming all currents leave supernode

$$-1 + \frac{V_1}{4} + \frac{V_1 - V_3}{1} + \frac{V_2}{1} - 2V_0 = 0$$
$$V_0 = V_1 - V_3$$
$$-4 + V_1 + 4V_1 - 4V_3 + 4V_2 - 8V_1 + 8V_3 = 0$$
$$-3V_1 + 4V_2 + 4V_3 = 4 \tag{15}$$

Assuming all currents leave V_3

$$2V_0 + \frac{V_3}{4} + \frac{V_3 - 10}{2} + \frac{V_3 - V_1}{1} = 0$$
$$8V_1 - 8V_3 + V_3 + 2V_3 - 20 + 4V_3 - 4V_1 = 0$$
$$4V_1 - V_3 = 20 \tag{16}$$

Solve using Cramer's rule

$$\begin{bmatrix} -3 & 4 & 4 \\ -1 & 1 & -1 \\ 4 & 0 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 20 \end{bmatrix}$$

Step 1: Determinant

$$\Delta = \begin{vmatrix} -3 & 4 & 4 \\ -1 & 1 & -1 \\ 4 & 0 & -1 \end{vmatrix} = -33$$

Step 2: Δ_1, Δ_2, Δ_3

$$\Delta_1 = \begin{vmatrix} 4 & 4 & 4 \\ 0 & 1 & -1 \\ 20 & 0 & -1 \end{vmatrix} = -164$$

$$\Delta_2 = \begin{vmatrix} -3 & 4 & 4 \\ -1 & 0 & -1 \\ 4 & 20 & -1 \end{vmatrix} = -160$$

$$\Delta_3 = \begin{vmatrix} -3 & 4 & 4 \\ -1 & 1 & 0 \\ 4 & 0 & 20 \end{vmatrix} = 4$$

Step 2: Solve for voltages

$$V_1 = \frac{\Delta_1}{\Delta} = -164/-33 = 4.97\ V$$

$$V_2 = \frac{\Delta_2}{\Delta} = -160/-33 = 4.848\ V$$

$$V_3 = \frac{\Delta_3}{\Delta} = 4/-33 = -0.12\ V$$